The if statement is more flexible than switch statement.

10.5 Iteration statements

An iteration (or looping) is a sequence of one or more statements that are repeatedly executed until a condition is satisfied. These statements are also called as control flow statements. It is used to reduce the length of code, to reduce time, to execute program and takes less memory space. C++ supports three types of iteration statements;

- for statement
- while statement
- do-while statement

All looping statements repeat a set statements as long as a specified condition is remains true. The specified condition is referred as a loop control. For all three loop statements, a true condition is any nonzero value and a zero value shows a false condition.

10.5.1 Parts of a loop

Every loop has four elements that are used for different purposes. These elements are

- Initialization expression
- Test expression
- Update expression
- The body of the loop

Initialization expression(s): The control variable(s) must be initialized before the control enters into loop. The initialization of the control variable takes place under the initialization expressions. The initialization expression is executed only once in the beginning of the loop.

Test Expression: The test expression is an expression or condition whose value decides whether the loop-body will be execute or not. If the expression evaluates to true (i.e., 1), the body of the loop gets executed, otherwise the loop is terminated.

In an entry-controlled loop, the test-expression is evaluated before the entering into a loop whereas in an exit-controlled loop, the test-expression is evaluated before exit from the loop.

Update expression: It is used to change the value of the loop variable. This statement is executed at the end of the loop after the body of the loop is executed.

The body of the loop: A statement or set of statements forms a body of the loop that are executed repetitively. In an entry-controlled loop, first the test-expression is evaluated and if it is nonzero, the body of the loop is executed otherwise the loop is terminated. In an exit-controlled loop, the body of the loop is executed first then the test-expression is evaluated. If the test-expression is true the body of the loop is repeated otherwise loop is terminated

10.5.2 for loop

The for loop is a entry- controlled loop and is the easiest looping statement which allows code to be executed repeatedly. It contains three different statements (initialization, condition or test-expression and update expression(s)) separated by semicolons.

The general syntax is:

```
for (initialization(s); test-expression; update expression(s))
{
        Statement 1;
        Statement 2;
        ..............
}
Statement-x;
```


Chapter 10 Page 152-179.indd 163 3/24/2020 9:21:33 AM